

PREPARATION LIST**Question 1**

Apparatus/ Reagents/Chemicals	Quantity (per student)
In a capped container, labelled C , 7% calcium chloride (dehydrate) solution, irritant	at least 50cm ³
In a beaker, labelled M , 10% dried milk solution	at least 100cm ³
In a capped container, labelled E , renin solution, irritant. See instructions for preparation for require concentration.	at least 10cm ³
In a beaker, labelled W , distilled water	at least 100cm ³
Boiling tube rack, suitable to hold 7 boiling tubes	1
Boiling tubes	7
Small beakers or containers to hold up to 50cm ³	6
Beaker to hold about 400cm ³ of water, labelled water-bath	1
Access to a sink and tap water	
Access to a source of hot water (at least 60°C)	
Beaker to hold about 200cm ³ of water, labelled cold water	1
Beaker to hold about 200cm ³ of water, labelled hot water	1
Thermometer, -10°C to 110°C	1
Paper towel	8
10cm ³ syringes	3
2cm ³ or 3cm ³ syringes	2
Black card, approximately 10cm x 10cm	1
Glass rod	1
Container labelled waste	1
Stopwatch	1
Glass marker pen	1
Safety goggles	1 pair

Question 2

Apparatus/ Material	Per Student
Microscopes (enough for ½ lab)	Instr to student: Shared per bench. Odd Seating index no. will use for the first 1hr 15 min followed by the even seating index no. the next 1 hr 15 min.
Slide labelled N1	

Question 3

Apparatus/Item	Per Student
Test tube with solution, labelled X	5cm ³
Test tube with solution, labelled Y	5cm ³
Test tube with solution, labelled Z	5cm ³
Dropper	Shared per bench
Beaker containing hydrogencarbonate indicator, labelled H	Shared per bench

Instr to students: Extra Reagent can be found on the teacher's bench. Raise your hands to ask for permission before coming to the front to take the reagent.

APPARATUS/ MATERIALS LIST (for students to check)**Question 1**

Apparatus/ Reagents/Chemicals	Quantity (per student)
In a capped container, labelled C , 20.0% calcium chloride solution, irritant	at least 50cm ³
In a beaker, labelled M , milk solution	at least 100cm ³
In a capped container, labelled E , 1.0% renin solution, irritant.	at least 10cm ³
In a beaker, labelled W , distilled water	at least 100cm ³
Boiling tube rack, suitable to hold 7 boiling tubes	1
Boiling tubes	7
Small beakers or containers to hold up to 50cm ³	6
Beaker to hold about 400cm ³ of water, labelled water-bath	1
Access to a sink and tap water	
Access to a source of hot water (Boiling Billy)	
Beaker to hold about 200cm ³ of water, labelled cold water (students to fill with tap water on their own)	1
Beaker to hold about 200cm ³ of water, labelled hot water (with hot water)	1
Thermometer, -10°C to 110°C	1
Paper towel	8
10cm ³ syringes	3
3cm ³ syringes	2
Black card, approximately 10cm x 10cm	1
Glass rod	1
Container labelled waste	1
Stopwatch	1
Glass marker pen	1
Safety goggles	1 pair

Instructions for preparation7% calcium chloride solution, labelled D1, at room temperature

This is prepared by adding 7 g of calcium chloride (dihydrate – $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$) to 70 cm³ of distilled water at room temperature. Make up to 100 cm³ with distilled water. Mix well.

This is **not** the same as the concentration given on the Question Paper. It is not necessary to inform the students that the concentration is different from that given in the Question Paper.

10% dried milk solution, labelled M

This is prepared by adding 10 g of dried milk powder to 80 cm³ of distilled water and mixing continuously until dissolved. Make up to 100 cm³ with distilled water.

The dried milk powder used must have at least 18 g protein in 100 g dried milk powder.

Rennin solution at room temperature, labelled E

Different sources of rennin have different activities. Before the examination, it is important to test the activity of the rennin and determine a suitable concentration to use. A 1% solution of rennin would be a suitable starting point.

In order to test the activity of the rennin solution, put 10 cm³ of milk, M, and 1 cm³ of calcium chloride solution, D1, into a boiling tube and mix well. Put into a water-bath at about 40 °C and leave for three minutes.

After three minutes, add 1 cm³ of rennin solution and gently swirl the boiling tube to mix. Start timing and rotate the boiling tube at an angle of about 45 degrees so that a film of milk forms as the boiling tube is rotated. Small clots of milk should appear between 20 and 45 seconds.

If the time is longer, then increase the concentration of the rennin solution. If the time is shorter, then decrease the concentration of the rennin solution.

It is not necessary to inform the students if the concentration of the enzyme is different from that given in the Question Paper.

Question 2

Apparatus/ Material	Per Student
Microscopes (enough for ½ lab)	Instr to student: Shared per bench. Odd Seating index no. will use for the first 1hr 15 min followed by the even seating index no. the next 1 hr 15 min.
Slide labelled N1	

Question 3

Apparatus/Item	Per Student
Test tube with solution, labelled X	5cm ³
Test tube with solution, labelled Y	5cm ³
Test tube with solution, labelled Z	5cm ³
Dropper	Shared per bench
Beaker containing hydrogencarbonate indicator, labelled H	Shared per bench

Instr to students: Extra Reagent can be found on the teacher's bench. Raise your hands to ask for permission before coming to the front to take the reagent. Students need not obtain permission to refill hot water from hot water access point.

Organisation of the examination

All candidates must have access to the materials required for Question 1 and 3 throughout the whole period of the examination

For Question 2, each candidate must have sole access to a microscope and slides for an uninterrupted period of 1 hour and 15 minutes only. Two candidates are not permitted to share the same microscope and the slide at the same time. Half of the candidates will have access to a microscope and slide for a period of 1 hour and 15 minutes from the start of the examinations. These candidates should be advised to start with Question 2. Once candidates have finished working with the microscope and the slide they can move on to

The other candidates should be advised to start with Question 1. After 1 hour 15 minutes, these candidates should be given access to the microscope and slide and advised to move on to Question 2 as soon as possible.

Candidates are advised to spend no more than:

- 60 minutes on Question 1
- 30 minutes on Question 2
- 55 minutes on Question 3

Access arrangement to microscopes, including advice on which question to start with and advisory timing must be explained to candidates before the start of the examination.